

Visión por Computador

Versión 2026



Transformaciones 2D

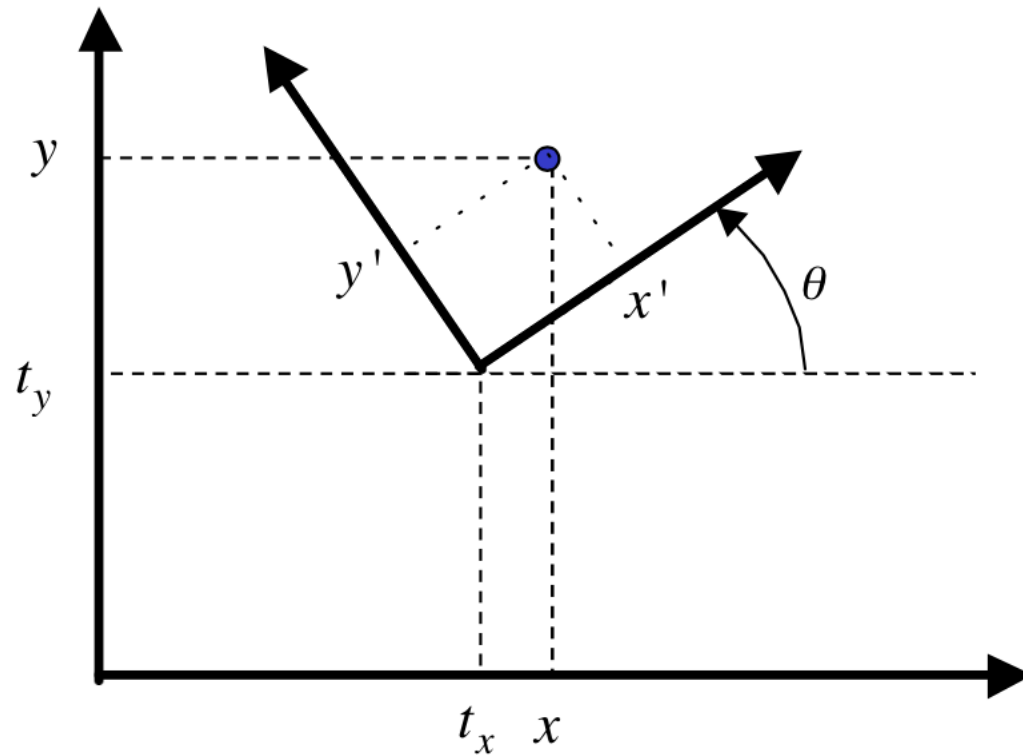
(CAPÍTULO II – PR02B)

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<http://domingomery.ing.uc.cl>

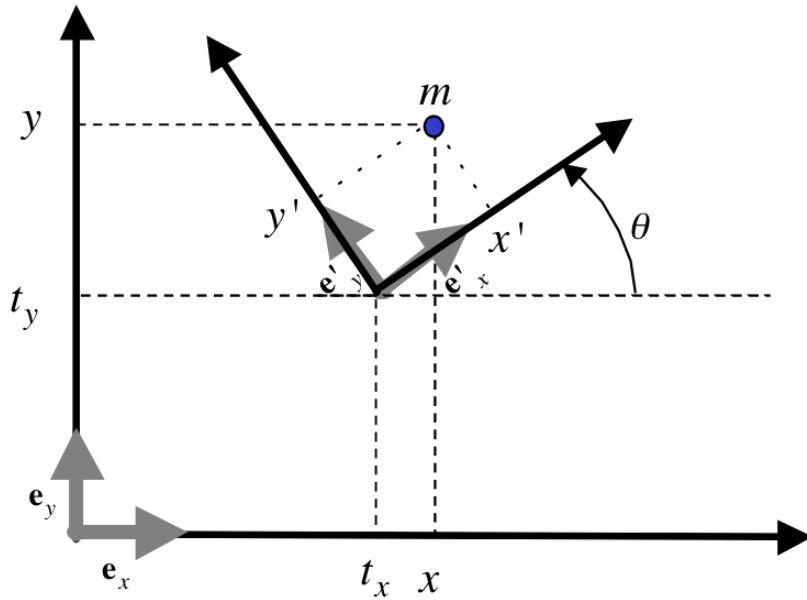


[Euclidean 2D Transformation]



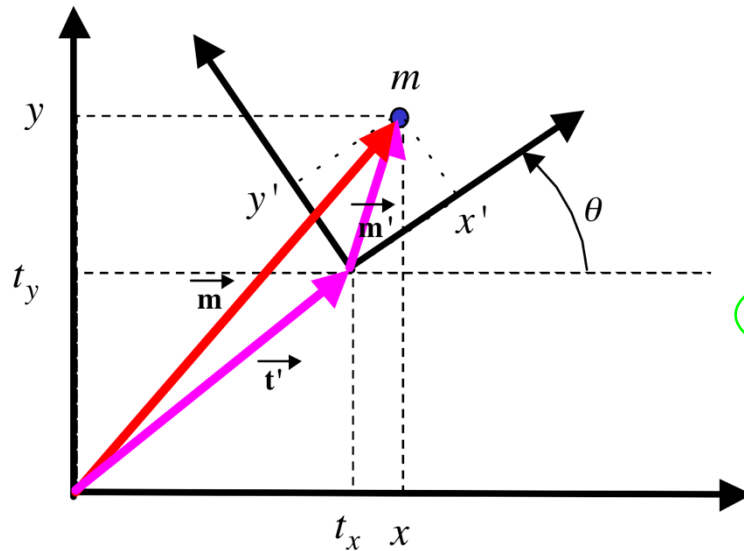
Problem: Given (x', y') and (t_x, t_y, θ) find (x, y)
or: Given (x, y) and (t_x, t_y, θ) find (x', y')

[Euclidean 2D Transformation]



$$\begin{aligned} \mathbf{e}'_x &= \cos(\theta)\mathbf{e}_x + \sin(\theta)\mathbf{e}_y \\ \mathbf{e}'_y &= -\sin(\theta)\mathbf{e}_x + \cos(\theta)\mathbf{e}_y \end{aligned} \quad \underbrace{x\mathbf{e}_x + y\mathbf{e}_y}_{\vec{\mathbf{m}}} = \underbrace{t_x\mathbf{e}_x + t_y\mathbf{e}_y}_{\vec{\mathbf{t}}'} + \underbrace{x'\mathbf{e}'_x + y'\mathbf{e}'_y}_{\vec{\mathbf{m}}'}$$

[Euclidean 2D Transformation]



$$\begin{aligned} \mathbf{e}'_x &= \cos(\theta)\mathbf{e}_x + \sin(\theta)\mathbf{e}_y \\ \mathbf{e}'_y &= -\sin(\theta)\mathbf{e}_x + \cos(\theta)\mathbf{e}_y \end{aligned}$$

$$\underbrace{x\mathbf{e}_x + y\mathbf{e}_y}_{\vec{m}} = \underbrace{t_x\mathbf{e}_x + t_y\mathbf{e}_y}_{\vec{t}'} + \underbrace{x'\mathbf{e}'_x}_{\vec{m}'_x} + \underbrace{y'\mathbf{e}'_y}_{\vec{m}'_y}$$

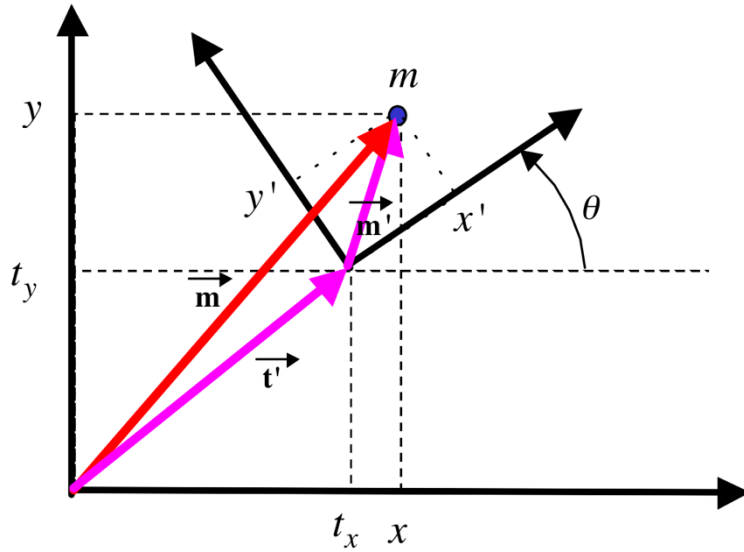
$$x\mathbf{e}_x + y\mathbf{e}_y = t_x\mathbf{e}_x + t_y\mathbf{e}_y + x'\cos(\theta)\mathbf{e}_x + x'\sin(\theta)\mathbf{e}_y - y'\sin(\theta)\mathbf{e}_x + y'\cos(\theta)\mathbf{e}_y$$

$$\mathbf{e}_x \text{ direction} \rightarrow x = x'\cos(\theta) - y'\sin(\theta) + t_x$$

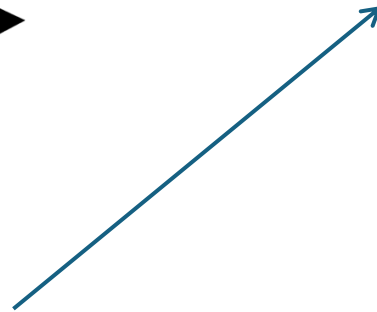
$$\mathbf{e}_y \text{ direction} \rightarrow y = x'\sin(\theta) + y'\cos(\theta) + t_y$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix} = \mathbf{R}' \begin{bmatrix} x' \\ y' \end{bmatrix} + \mathbf{t}'$$

[Euclidean 2D Transformation]

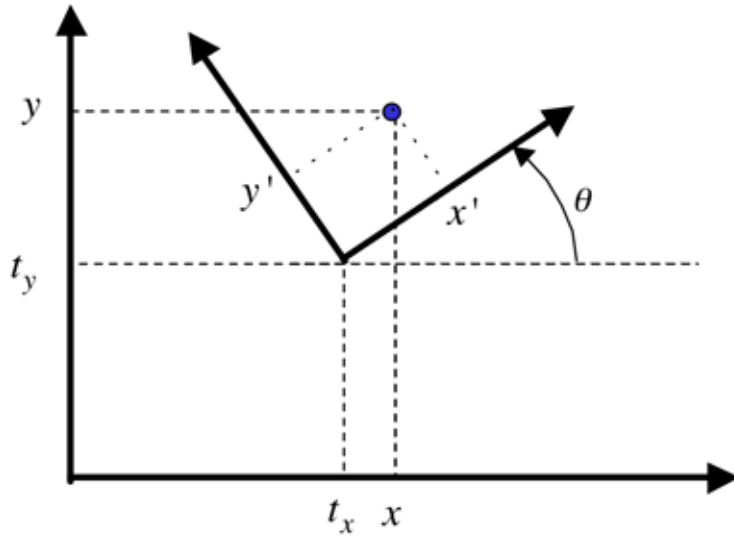


$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}' & \mathbf{t}' \\ \mathbf{0}^\top & 1 \end{bmatrix} \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix}$$



$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix} = \mathbf{R}' \begin{bmatrix} x' \\ y' \end{bmatrix} + \mathbf{t}'$$

[Euclidean 2D Transformation]

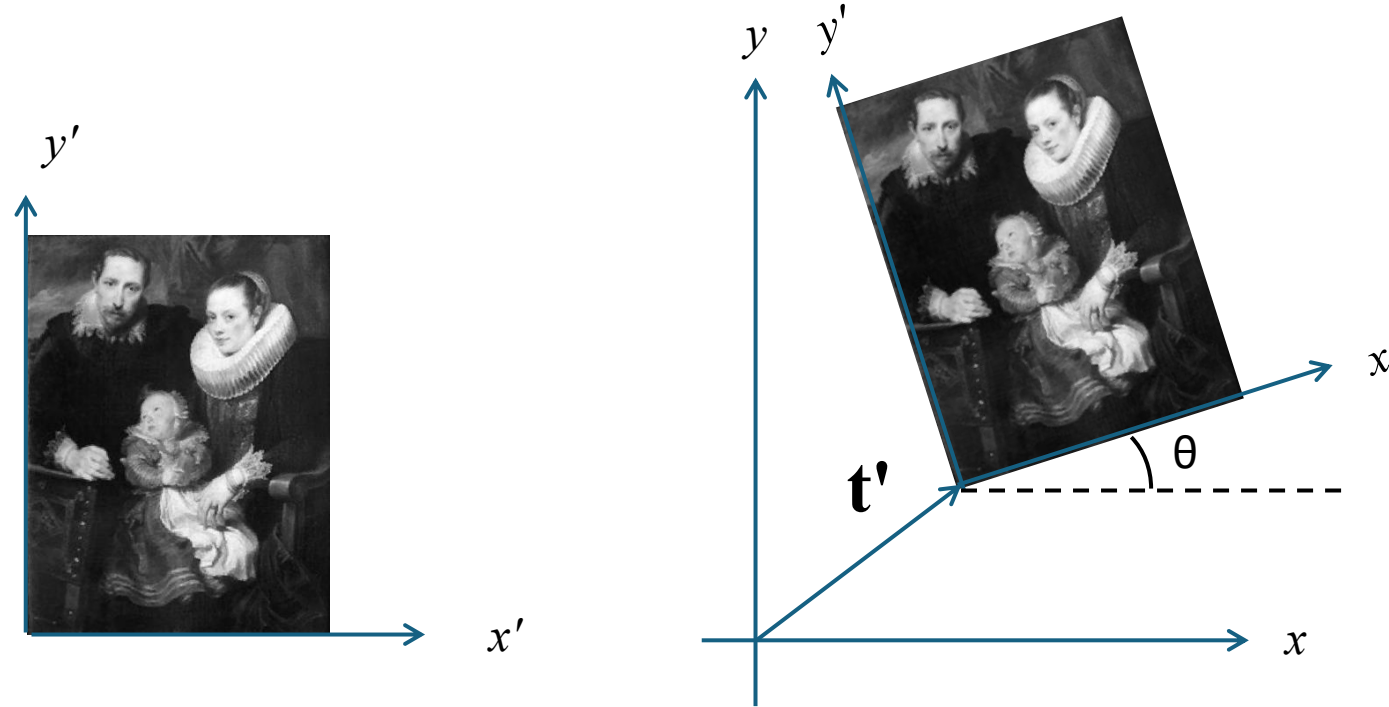


$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}' & \mathbf{t}' \\ \mathbf{0}^\top & 1 \end{bmatrix} \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix}$$

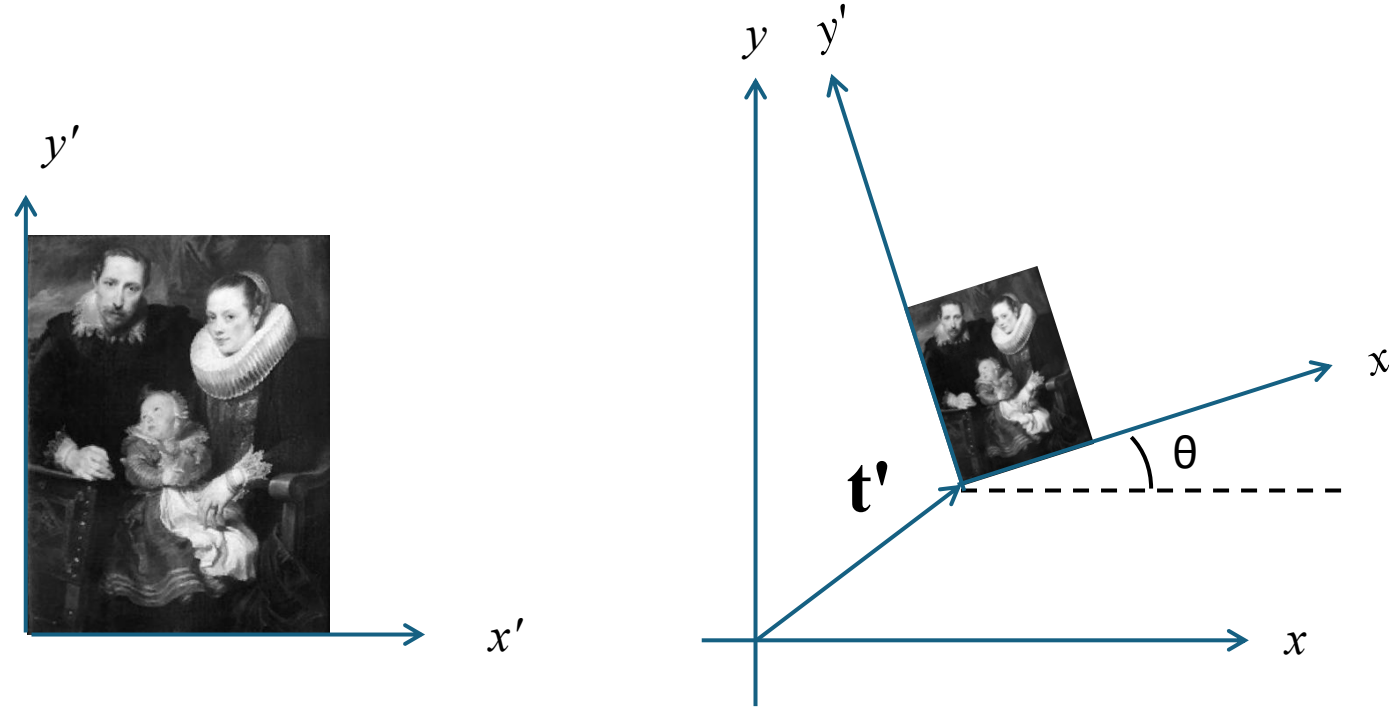
$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}' & \mathbf{t}' \\ \mathbf{0}^\top & 1 \end{bmatrix}^{-1} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Problem: Given (x', y') and (t_x, t_y, θ) find (x, y)
or: Given (x, y) and (t_x, t_y, θ) find (x', y')

[Euclidean 2D Transformation]



[Similarity 2D Transformation]



$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} s\mathbf{R}' & \mathbf{t}' \\ \mathbf{0}^\top & 1 \end{bmatrix} \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix}$$

[2D Transformations – H matrix]



$$\begin{bmatrix} \cos(\theta) & -\sin(\theta) & t_x \\ \sin(\theta) & \cos(\theta) & t_y \\ 0 & 0 & 1 \end{bmatrix}$$

[2D Transformations – H matrix]

Euclidean



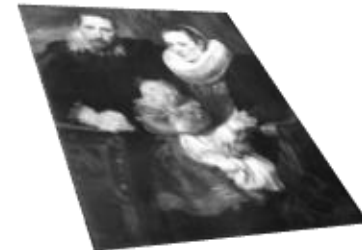
Similarity



Affine



General



$$\begin{bmatrix} \cos(\theta) & -\sin(\theta) & t_x \\ \sin(\theta) & \cos(\theta) & t_y \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} s \cos(\theta) & -s \sin(\theta) & t_x \\ s \sin(\theta) & s \cos(\theta) & t_y \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix}$$

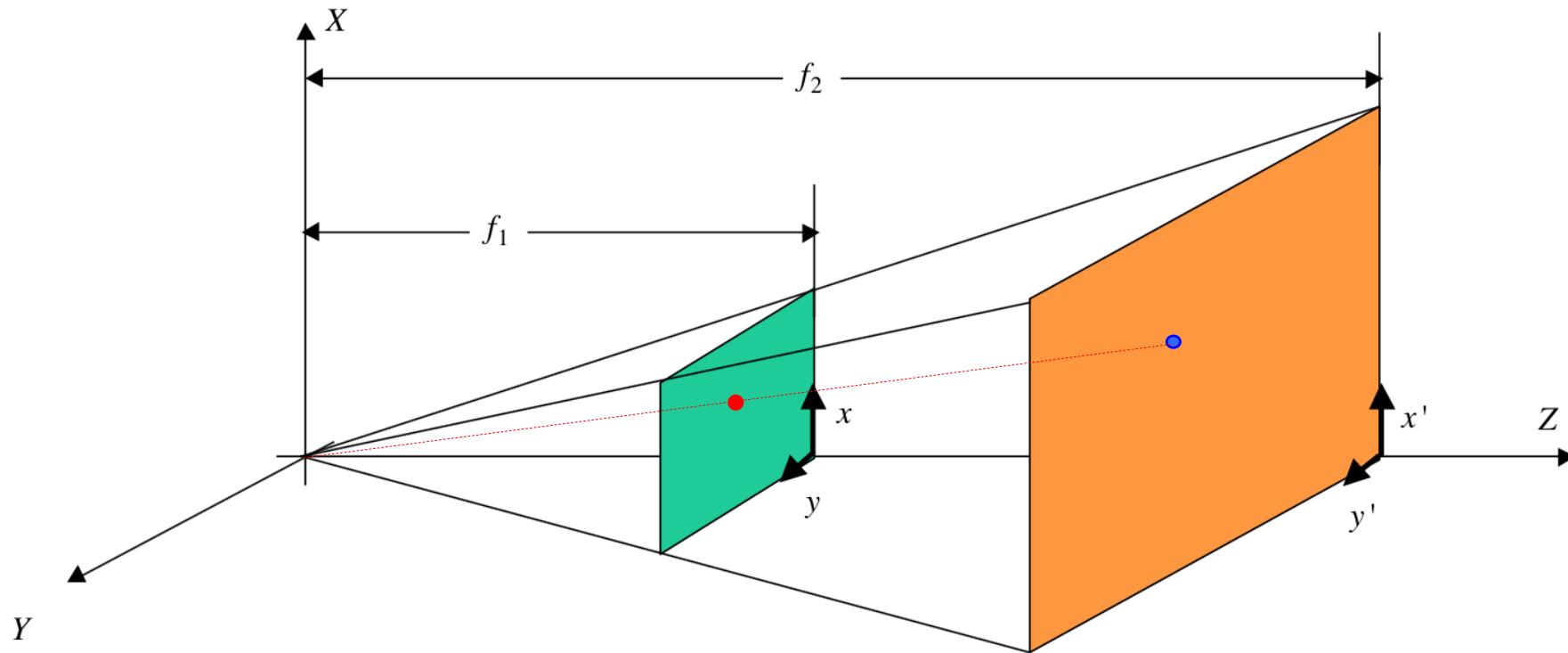
[Homography]

Transformation from two 2D spaces, where straight lines are transformed as straight lines.

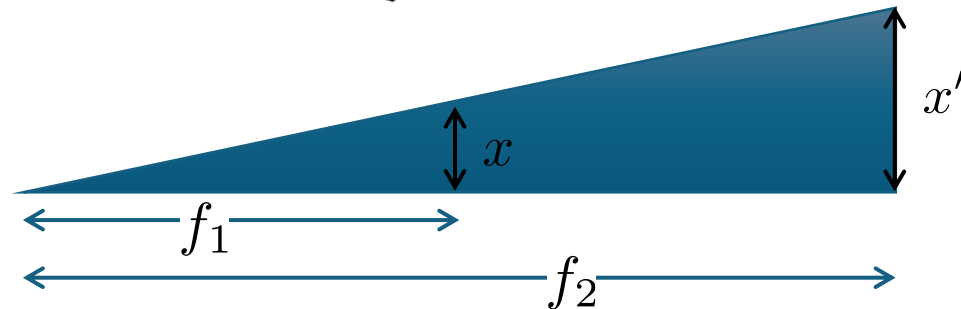


[Homography: Example]

Problem: Given $\bullet (x, y)$, find $\bullet (x', y')$
(planes and are parallel and perpendicular to Z).



$$\frac{x'}{f_2} = \frac{x}{f_1} \quad \frac{y'}{f_2} = \frac{y}{f_1}$$



[Homography: Example]

$$\frac{x'}{f_2} = \frac{x}{f_1} \quad \frac{y'}{f_2} = \frac{y}{f_1}$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} f_2/f_1 & 0 & 0 \\ 0 & f_2/f_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

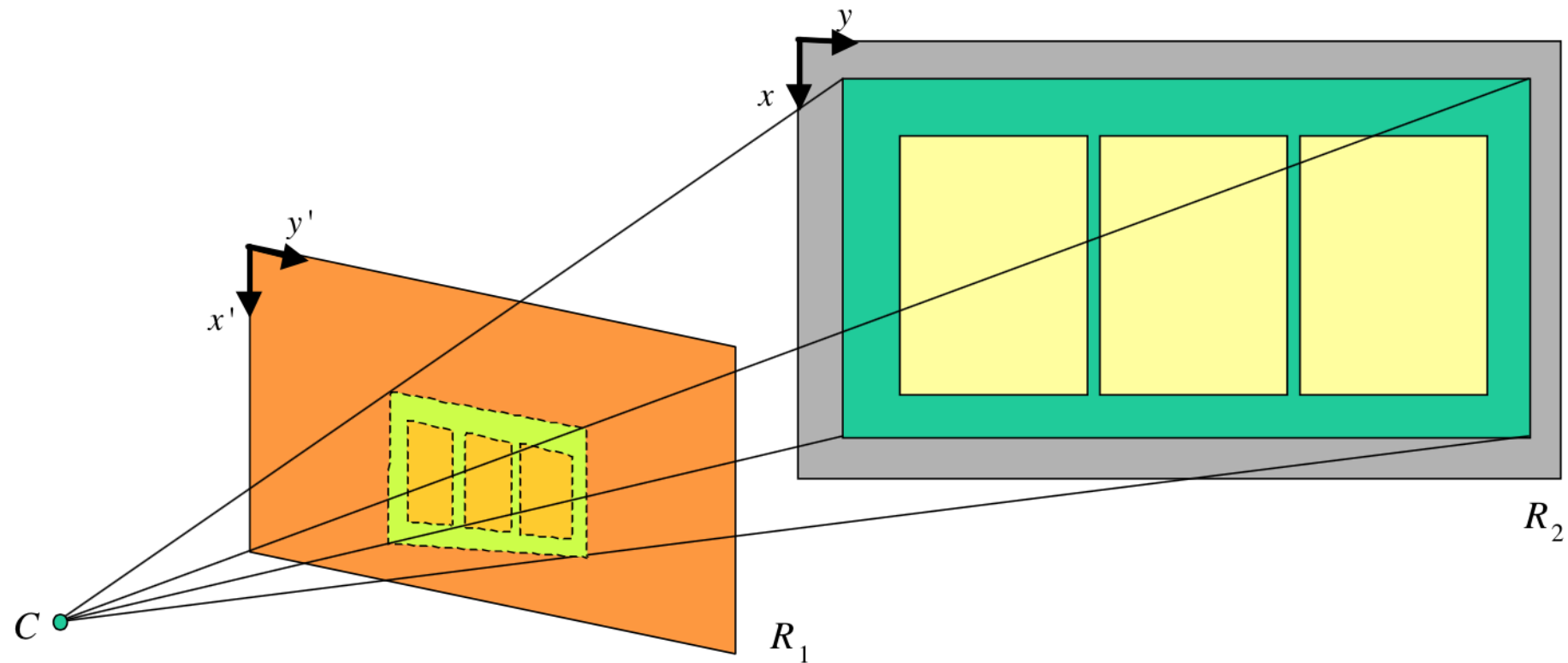
$$\mathbf{m}' = \mathbf{H}\mathbf{m} \quad \mathbf{H} = \begin{bmatrix} f_2/f_1 & 0 & 0 \\ 0 & f_2/f_1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

[Homography]

Transformation from two 2D spaces, where straight lines are transformed as straight lines.



[Homography: Planes R_1 and R_2 are not parallel]

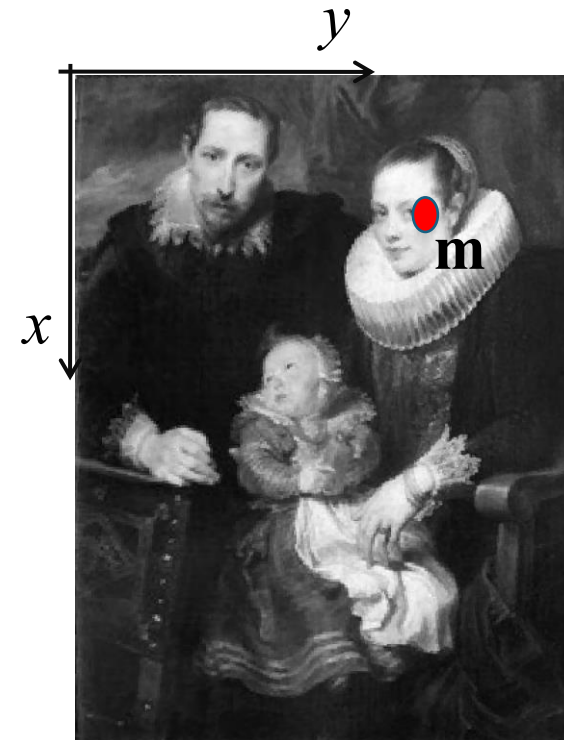
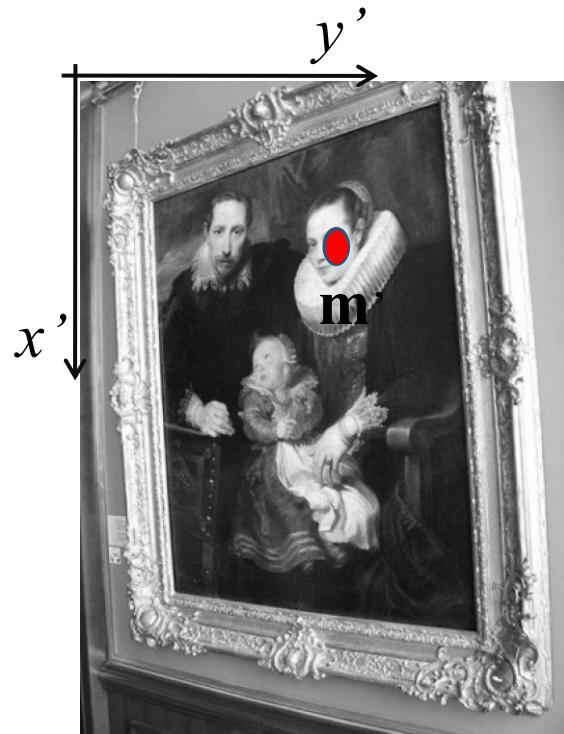


$$\lambda \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} x'_1 \\ x'_2 \\ x'_3 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

[Homography]

Demo: pág 21 (D.Mery: Apuntes de Visión por Computador)

$$\lambda \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} x_1' \\ x_2' \\ x_3' \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$



$$\mathbf{m}' = \mathbf{H}\mathbf{m}$$

[Homography]

$$\lambda \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} x_1' \\ x_2' \\ x_3' \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$



$$x' = \frac{x_1'}{x_3'} = \frac{h_{11}x + h_{12}y + h_{13}}{h_{31}x + h_{32}y + h_{33}}$$

$$y' = \frac{x_2'}{x_3'} = \frac{h_{21}x + h_{22}y + h_{23}}{h_{31}x + h_{32}y + h_{33}}$$

[Homography]

$$x' = \frac{x'_1}{x'_3} = \frac{h_{11}x + h_{12}y + h_{13}}{h_{31}x + h_{32}y + h_{33}}$$

$$y' = \frac{x'_2}{x'_3} = \frac{h_{21}x + h_{22}y + h_{23}}{h_{31}x + h_{32}y + h_{33}}$$

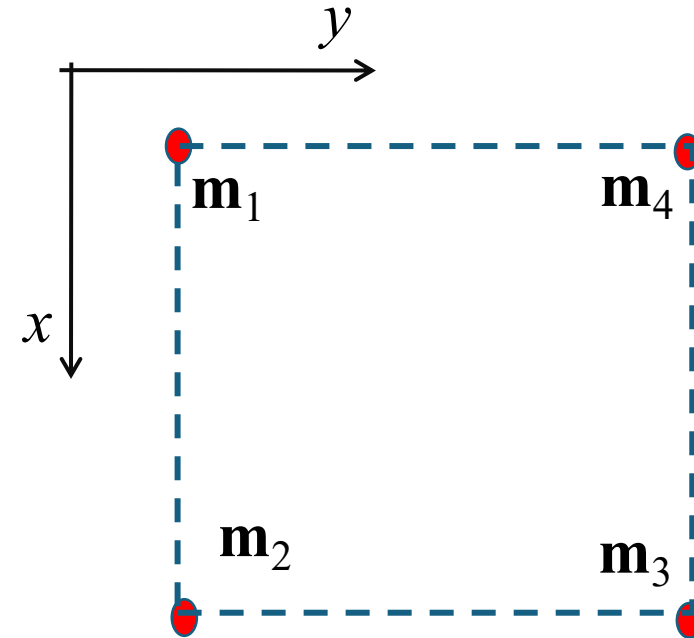
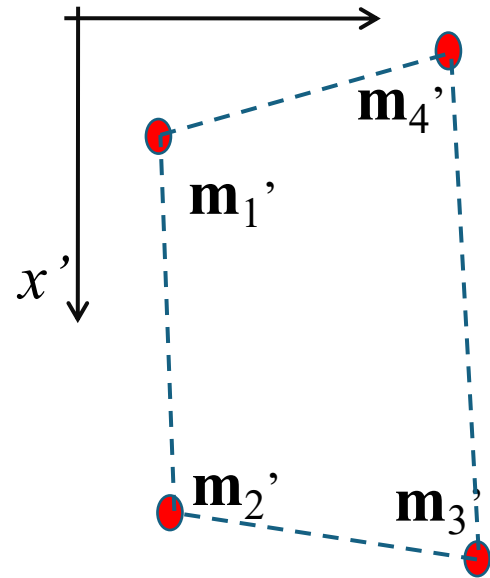
with $h_{33} = 1$

$$\underbrace{\begin{bmatrix} x & y & 1 & 0 & 0 & 0 & -x'x & -x'y \\ 0 & 0 & 0 & x & y & 1 & -y'x & -y'y \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} h_{11} \\ h_{12} \\ h_{13} \\ h_{21} \\ h_{22} \\ h_{23} \\ h_{31} \\ h_{32} \end{bmatrix}}_{\mathbf{h}} = \underbrace{\begin{bmatrix} x' \\ y' \end{bmatrix}}_{\mathbf{b}}$$

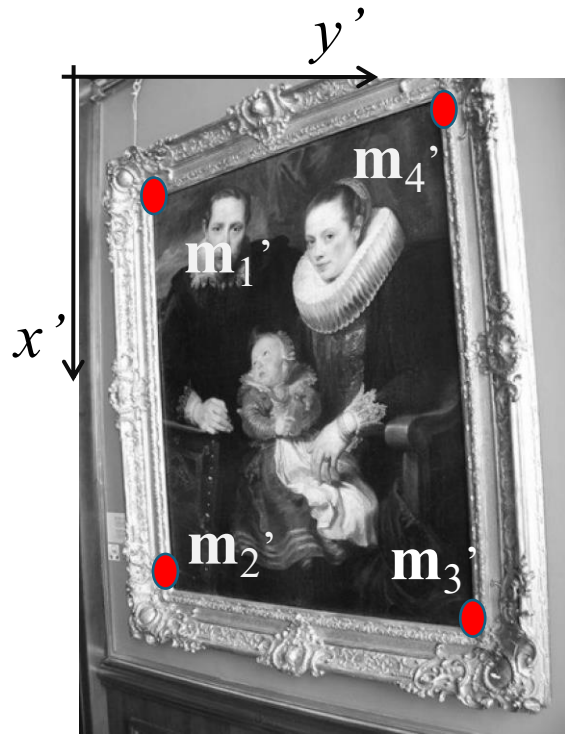
$\mathbf{A}\mathbf{h} = \mathbf{b}$

[Homography]

y'



[Homography]



$$\tilde{\mathbf{A}} \begin{bmatrix} \mathbf{A}_1 \\ \mathbf{A}_2 \\ \vdots \\ \mathbf{A}_n \end{bmatrix} \mathbf{h} = \begin{bmatrix} \mathbf{b}_1 \\ \mathbf{b}_2 \\ \vdots \\ \mathbf{b}_n \end{bmatrix}$$

$$\mathbf{h} = \tilde{\mathbf{A}}^{-1} \mathbf{b} \quad \text{for } n = 4$$

$$\mathbf{h} = [\tilde{\mathbf{A}}^T \tilde{\mathbf{A}}]^{-1} \tilde{\mathbf{A}}^T \mathbf{b} \quad \text{for } n > 4$$

Implementation

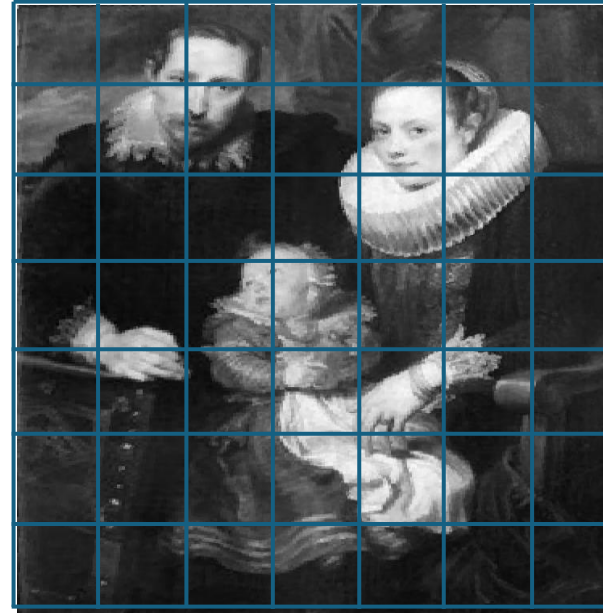
SOURCE

TARGET

I'



I

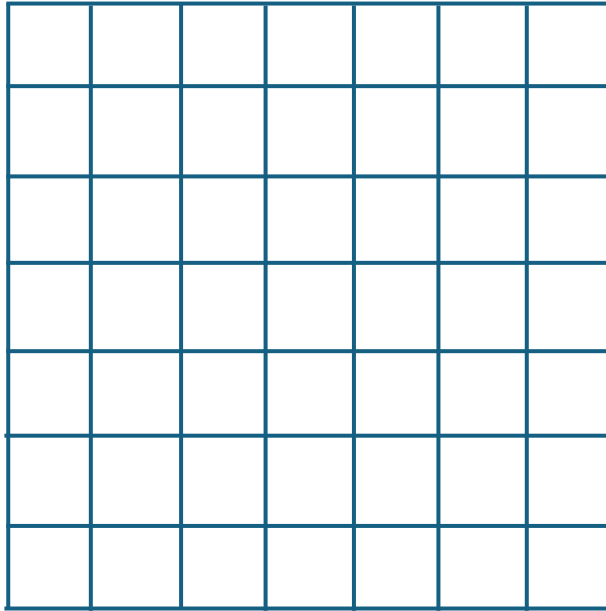


H
←

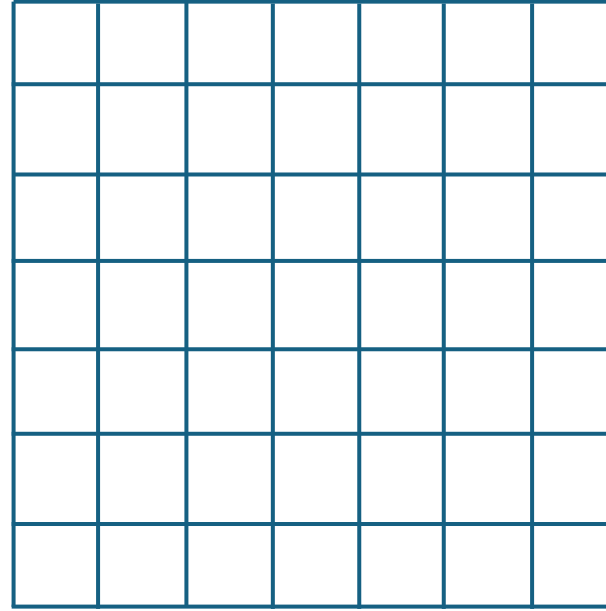
SOURCE

TARGET

I'



I

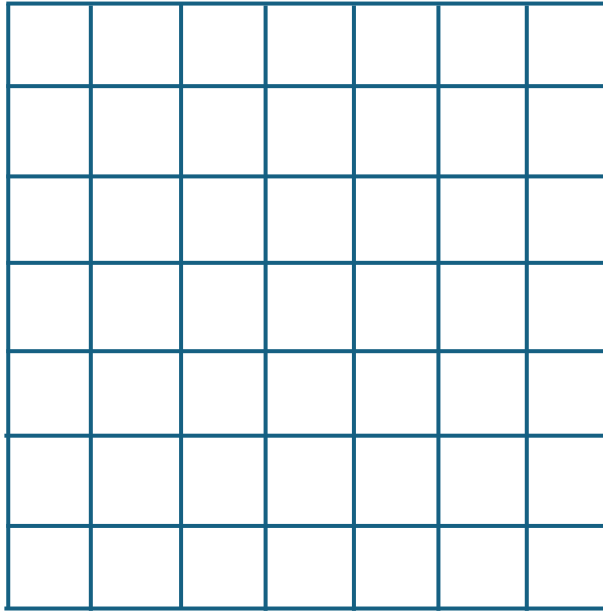


H
←

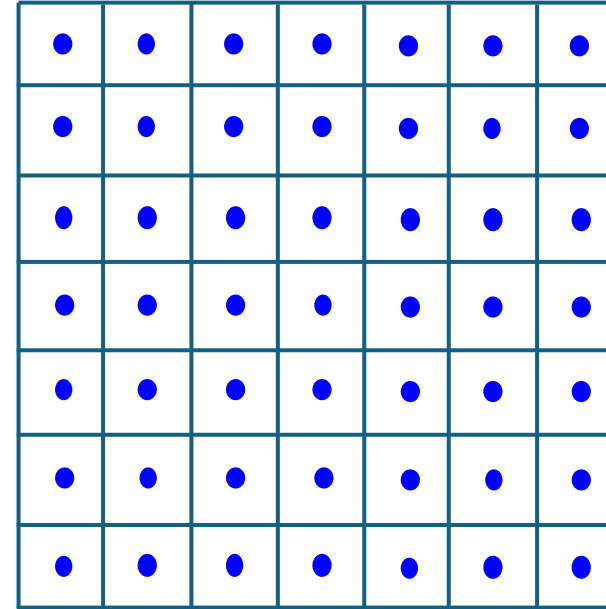
SOURCE

TARGET

I'



I



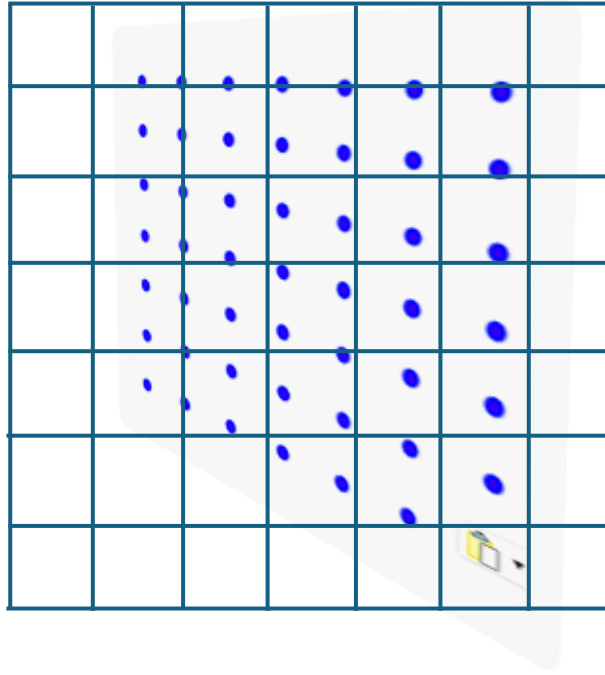
H
←

ALGORITHM: For each pixel in TARGET

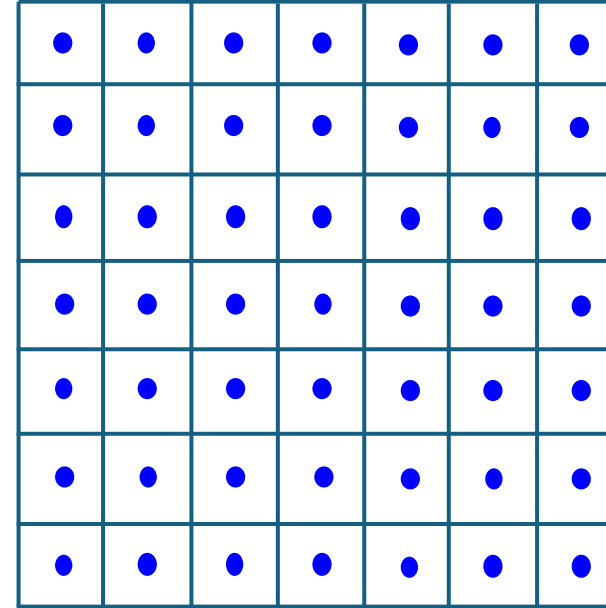
SOURCE

TARGET

I'



I



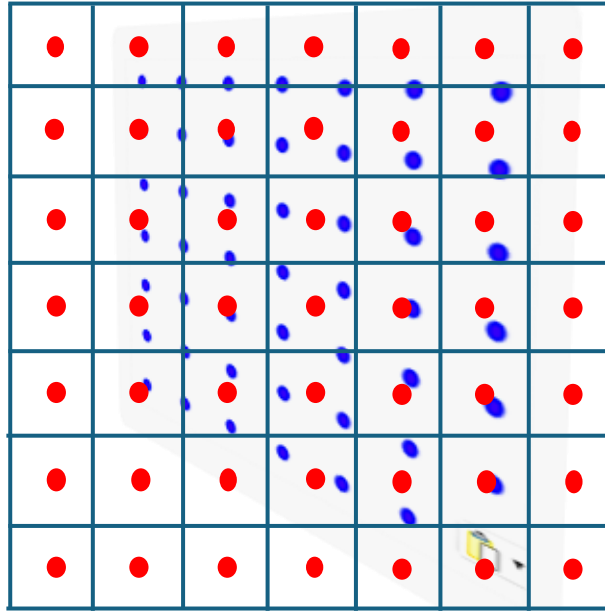
H
←

ALGORITHM: For each pixel in TARGET
- compute the corresponding position in SOURCE

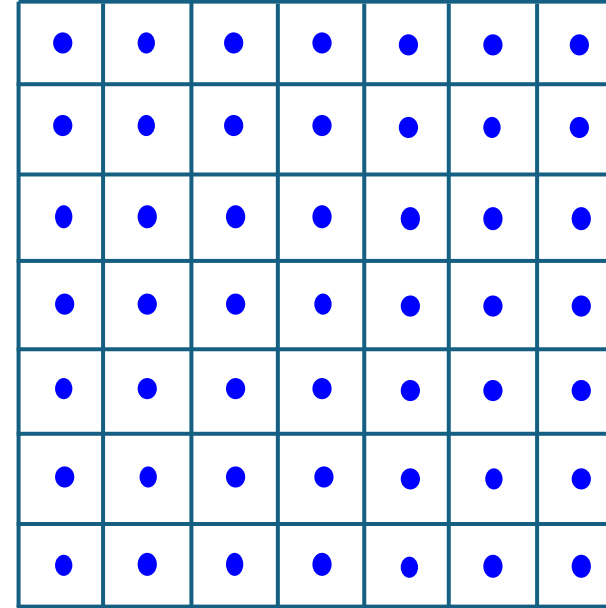
SOURCE

TARGET

I'



I

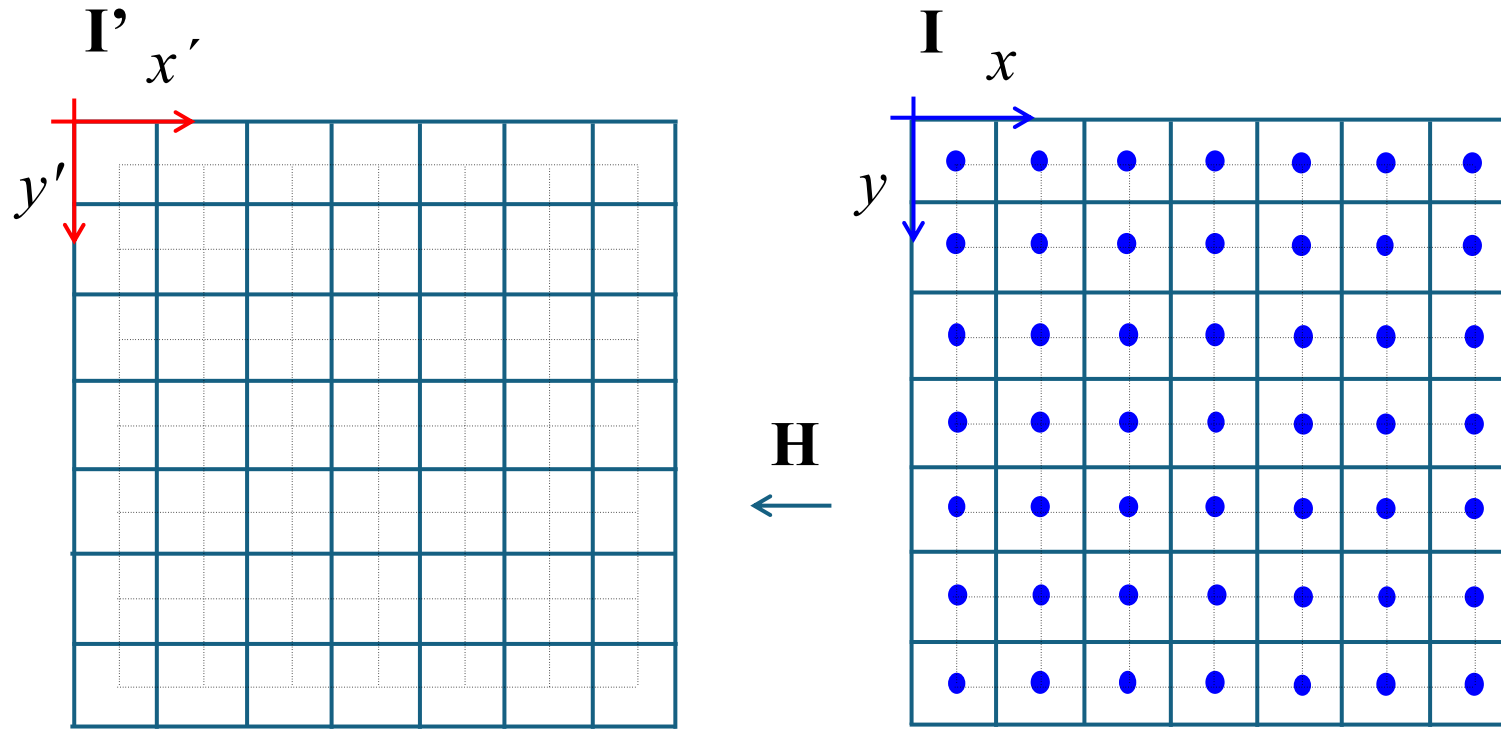


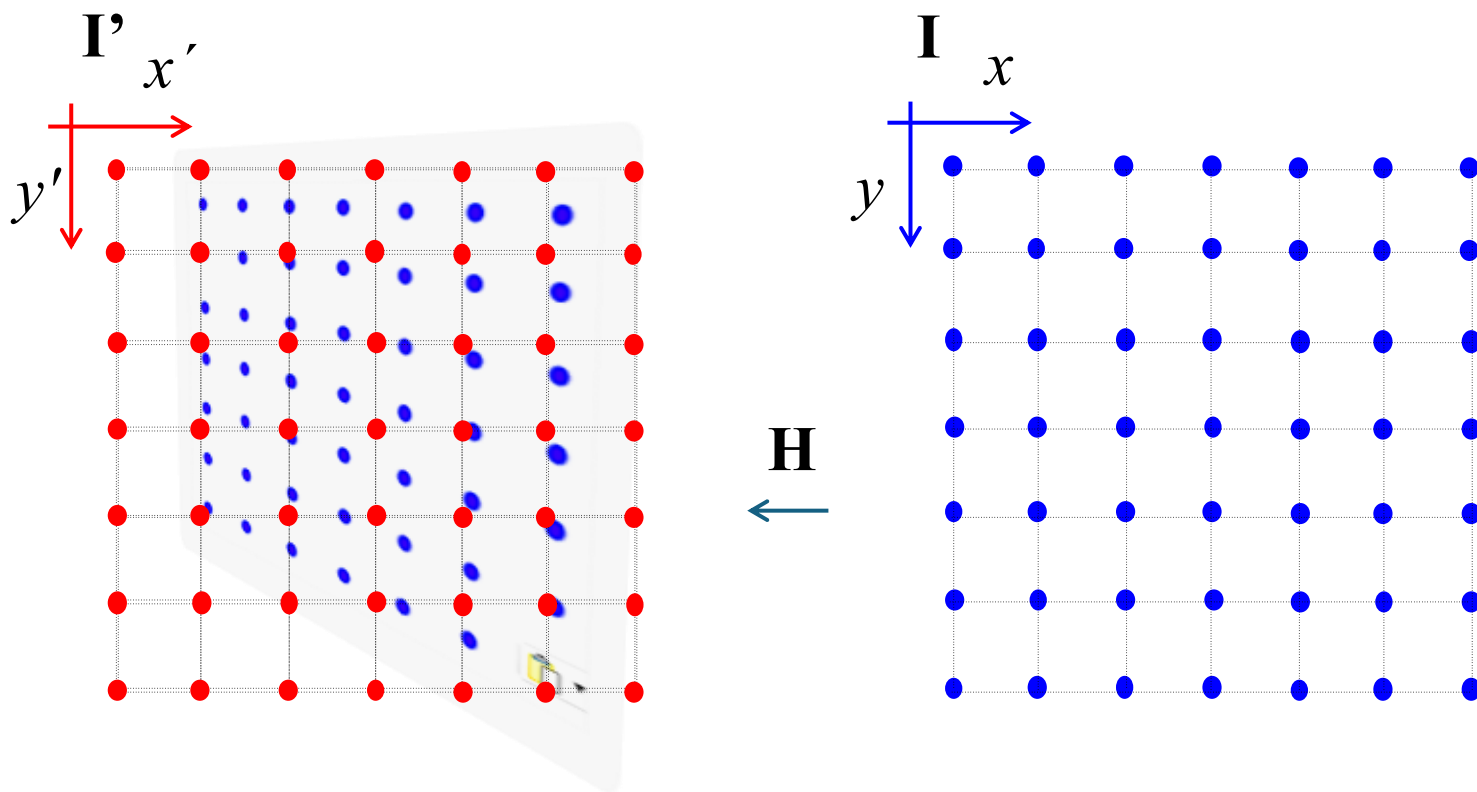
H
←

ALGORITHM:

For each pixel in TARGET

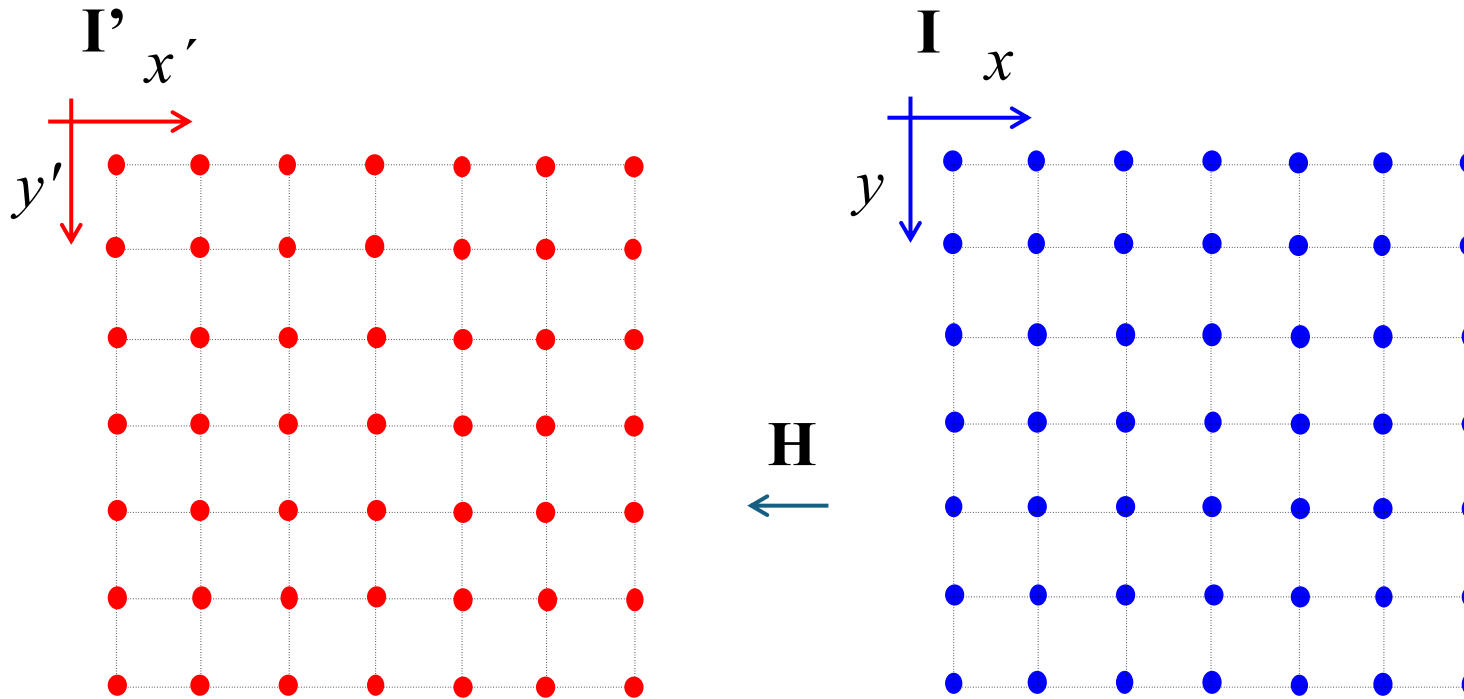
- compute the corresponding position in SOURCE
- interpolate the grayvalue using SOURCE pixels.





Coordinate Transformation:

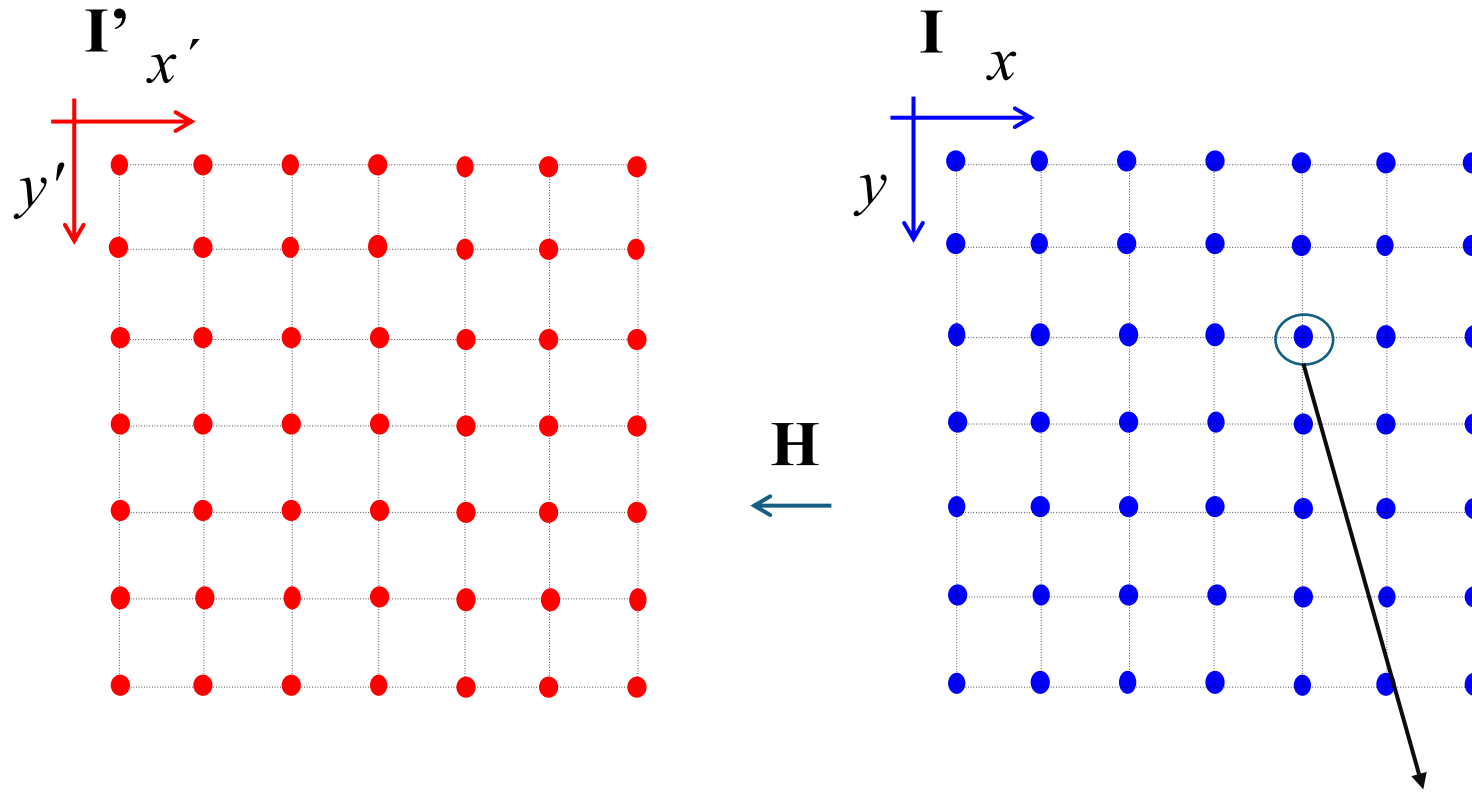
$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} x_1' \\ x_2' \\ x_3' \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$



Coordinate Transformation:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} x_1' \\ x_2' \\ x_3' \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

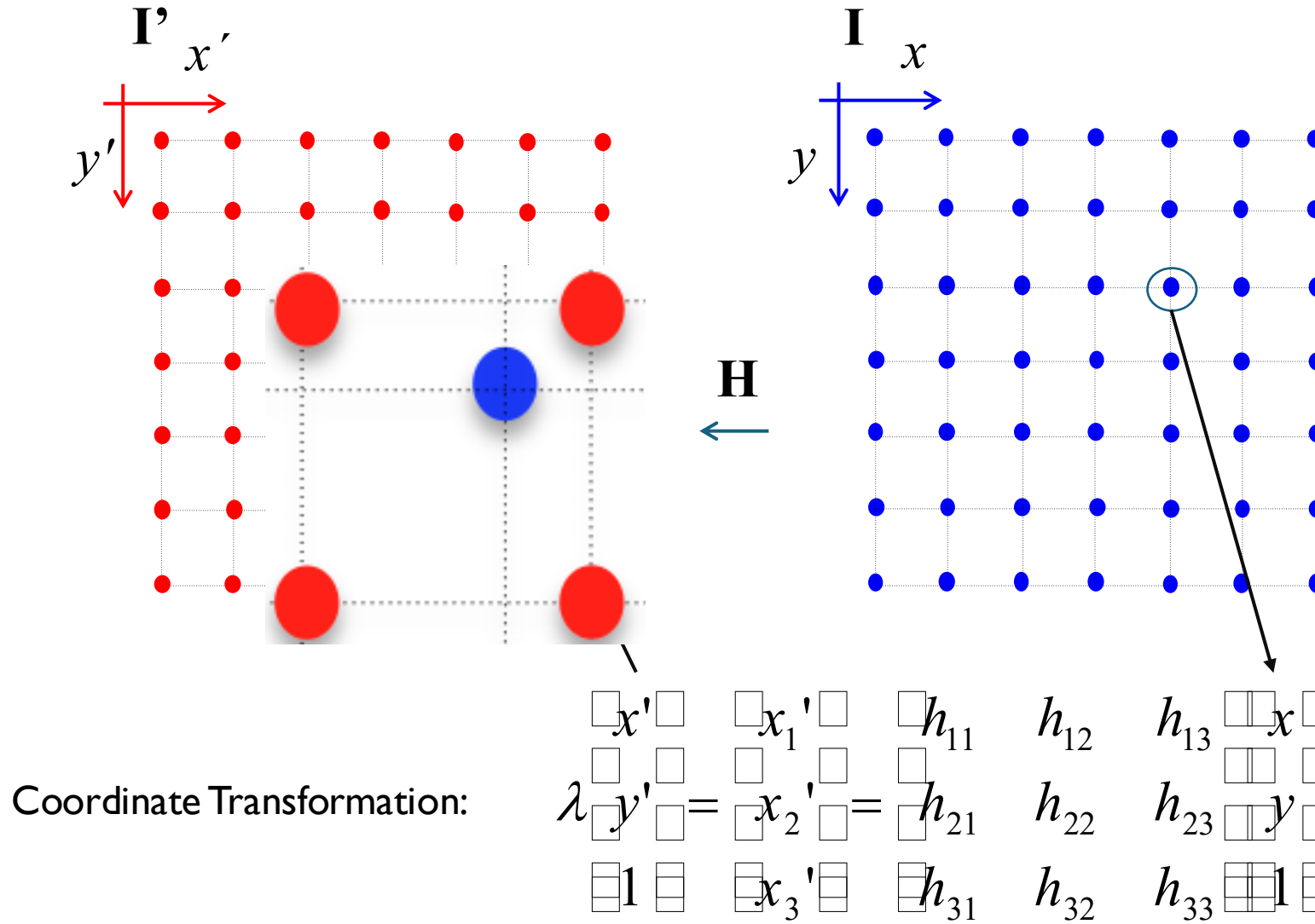
EXAMPLE

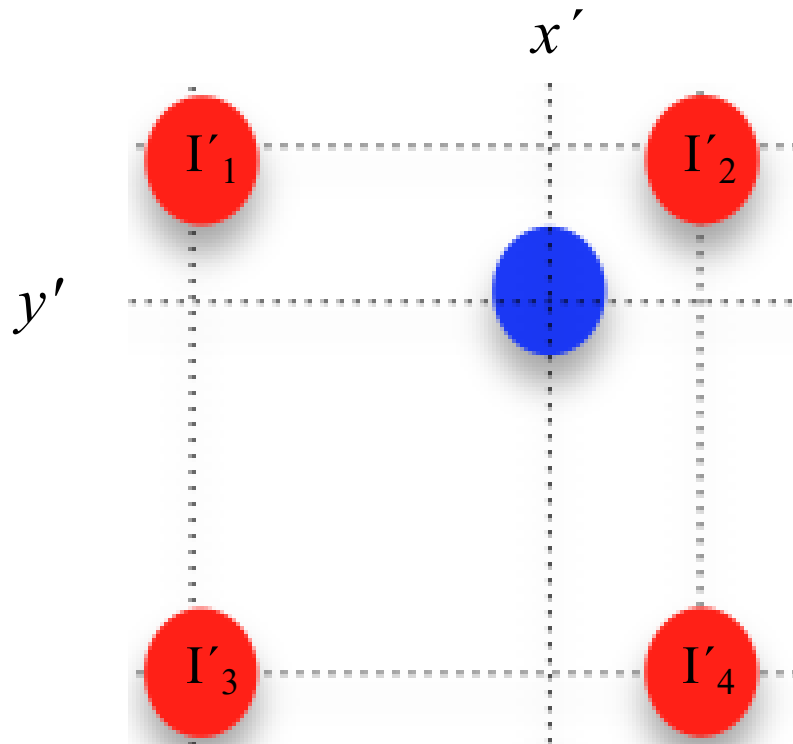


Coordinate Transformation:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} x_1' \\ x_2' \\ x_3' \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

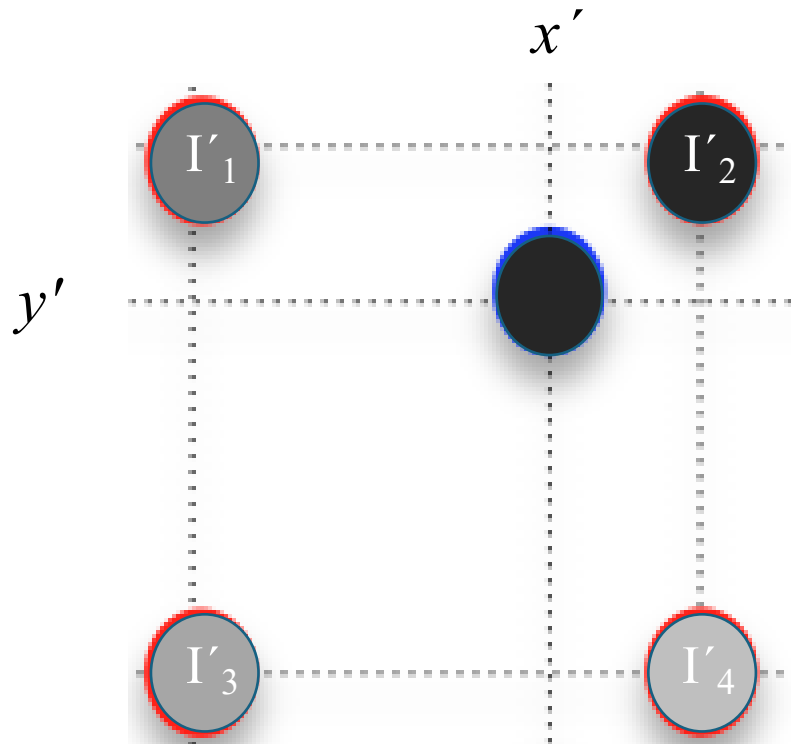
EXAMPLE





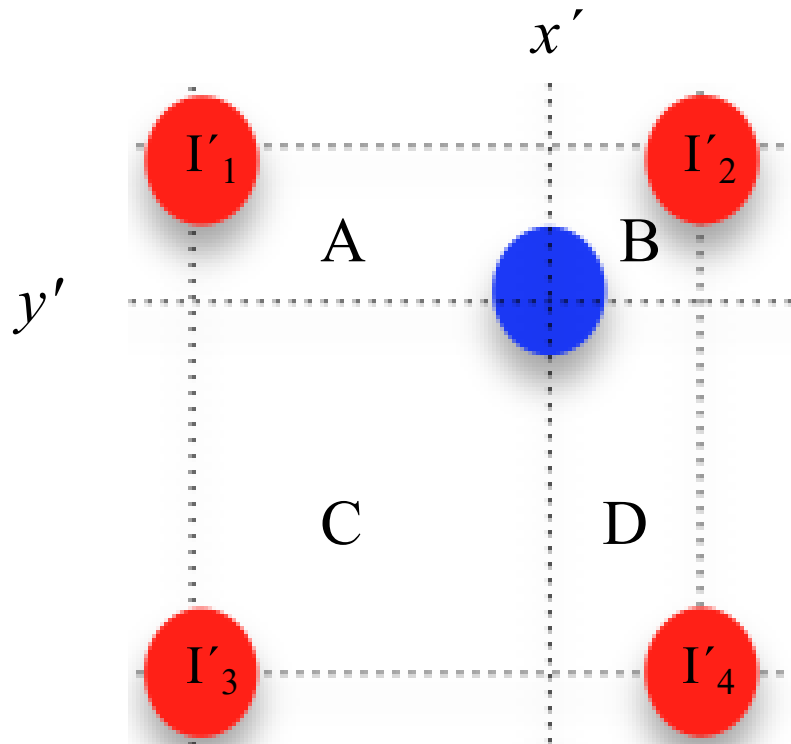
Nearest pixel

$$I(x,y) = \text{interpolation } \{I'(x',y')\} = I_2$$



Nearest pixel

$$I(x,y) = \text{interpolation } \{I'(x',y')\} = I_2$$

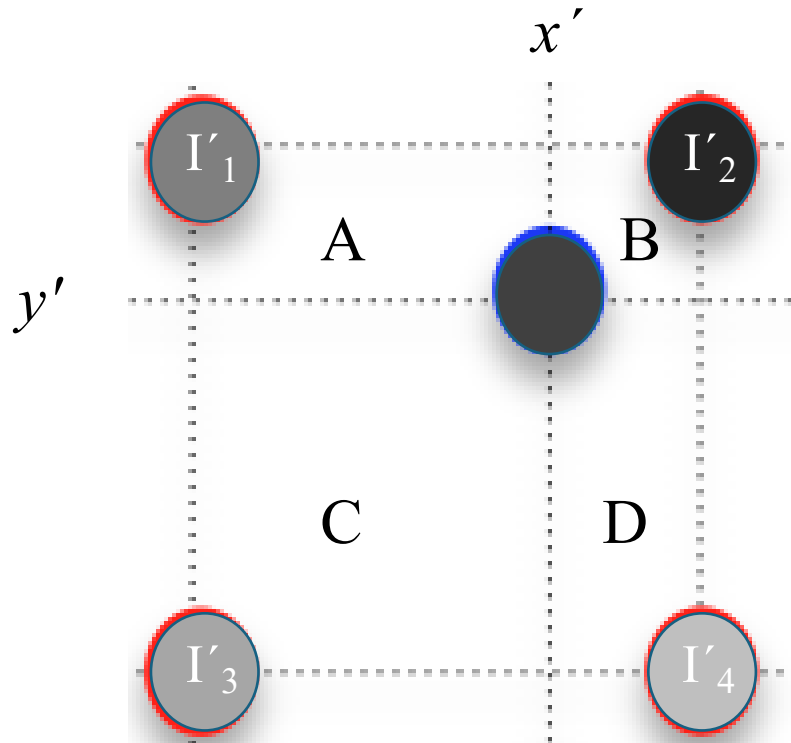


Bilinear interpolation

$I(x,y) = \text{interpolation } \{I'(x',y')\} =$

$$AI'_4 + BI'_3 + CI'_2 + DI'_1$$

$$A+B+C+D = 1$$



Bilinear interpolation

$I(x,y) = \text{interpolation } \{I'(x',y')\} =$

$$AI'_4 + BI'_3 + CI'_2 + DI'_1$$

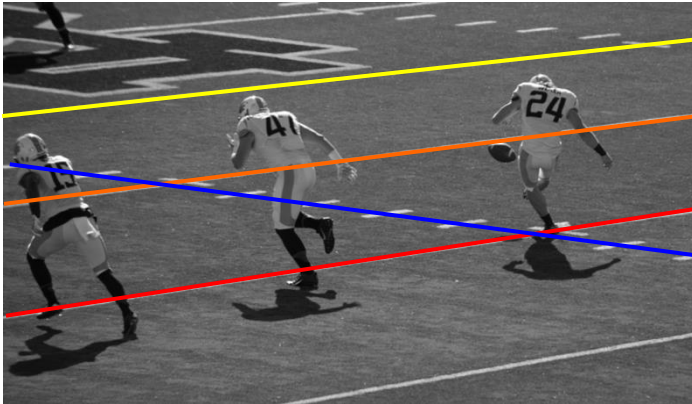
$$A+B+C+D = 1$$

Implementation:

1. $x = 1, \dots, N$; $y = 1, \dots, M$:
2. $\mathbf{m} = [x \ y \ 1]^\top$.
3. $\mathbf{m}' = \mathbf{H}\mathbf{m}$.
4. $x' = m_1/m_3$ y $y' = m_2/m_3$.
5. $\tilde{x} = \text{fix}(x')$, $\tilde{y} = \text{fix}(y')$, $\Delta x' = x' - \tilde{x}$ y $\Delta y' = y' - \tilde{y}$.
6.
$$I(x, y) = [I'(\tilde{x}+1, \tilde{y}) - I'(\tilde{x}, \tilde{y})]\Delta x' + [I'(\tilde{x}, \tilde{y}+1) - I'(\tilde{x}, \tilde{y})]\Delta y' + [I'(\tilde{x}+1, \tilde{y}+1) + I'(\tilde{x}, \tilde{y}) - I'(\tilde{x}+1, \tilde{y}) - I'(\tilde{x}, \tilde{y}+1)]\Delta x'\Delta y' + I'(\tilde{x}, \tilde{y})$$

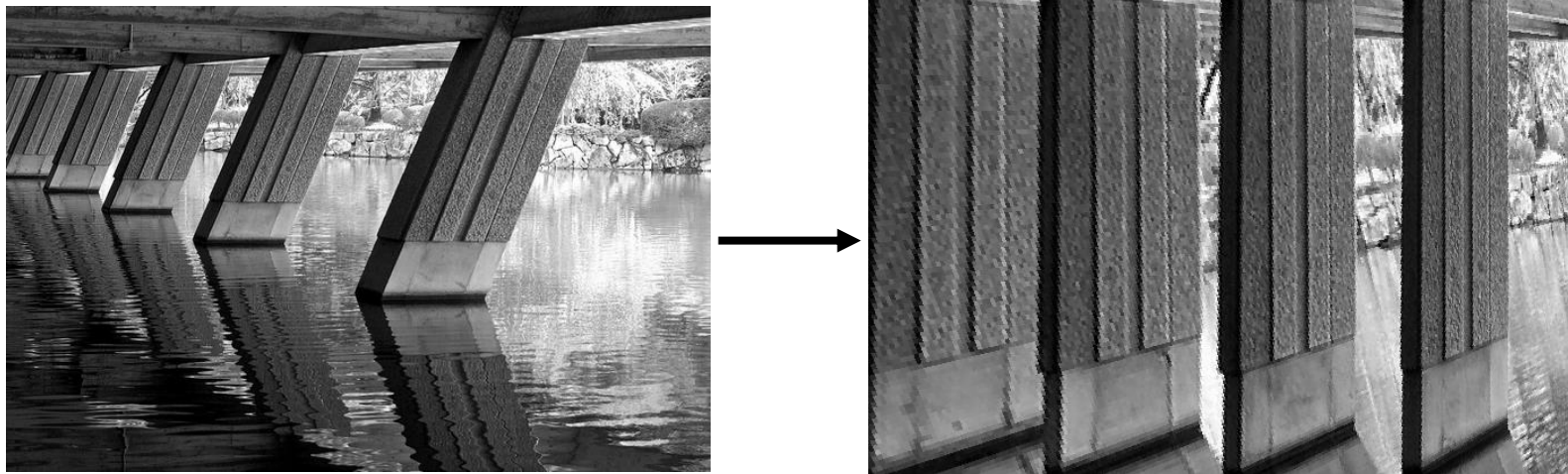
[Homography]

More examples.



[Homography]

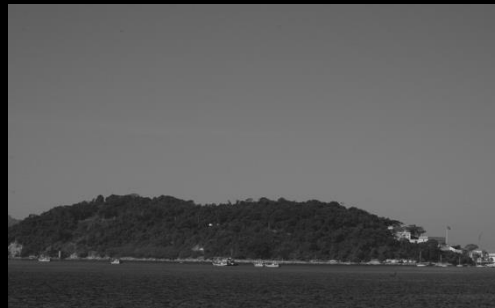
More examples.



Mosaic Example I

Rio de Janeiro

















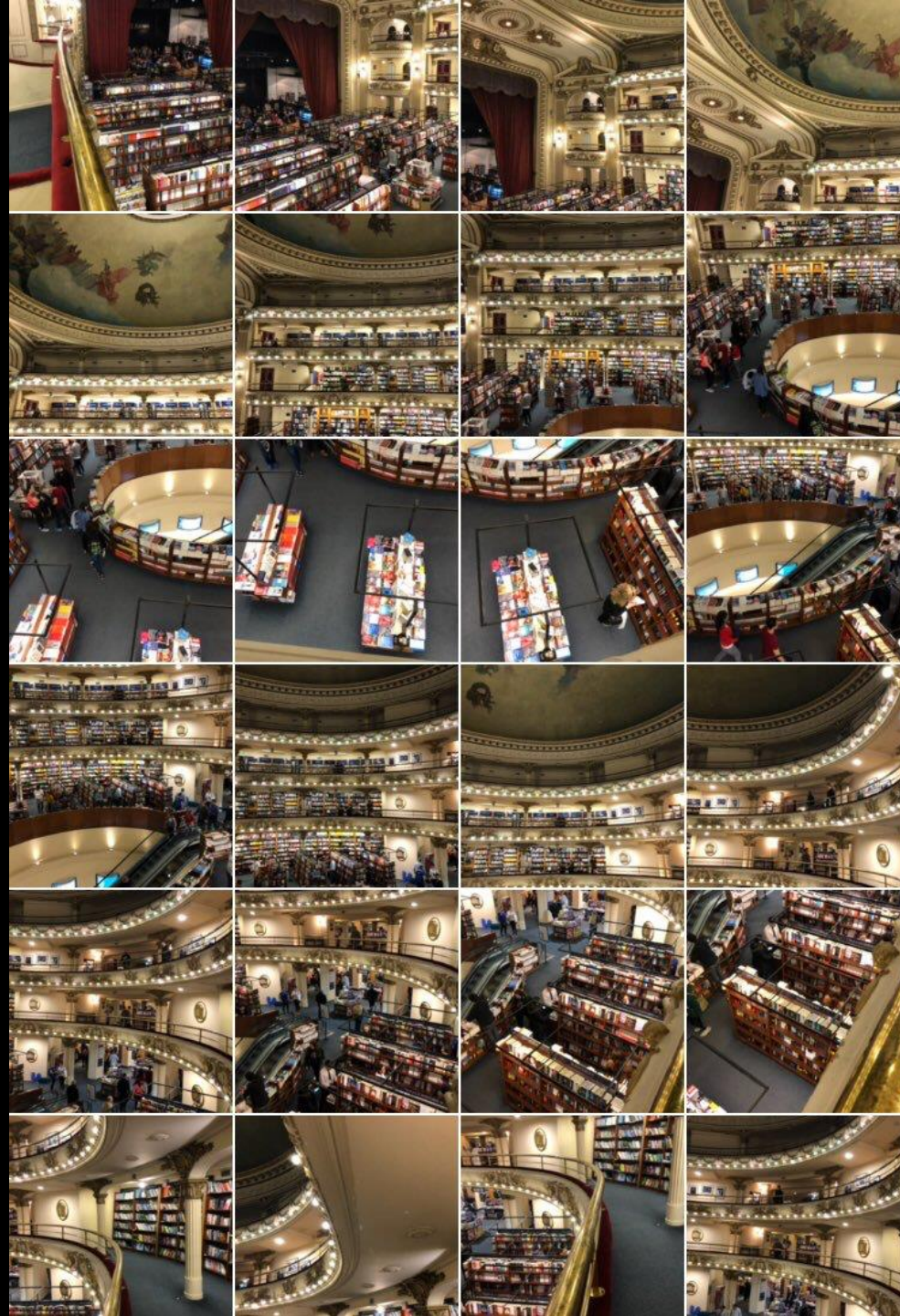






Mosaic Example 2

Buenos Aires
Librería Ateneo





Mosaic Example 3

Horseshoe Bend
Arizona

